DUJARDIN JULIEN

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Education

Mines Saint-Étienne, Saint-Étienne, France

Master of Engineering (M.Eng.) in Embedded Systems (Diplôme d'Ingénieur) **10/2023 – 09/2026**

- Research Project / Master's Thesis: "Comparative analysis of energy efficiency: Implementing Neural Networks on FPGA vs. CPU architectures."
- Core Modules: Microelectronics, Real-time Systems, FPGA Design, Signal Processing, High-frequency circuit design.

Université Picardie Jules Verne, Amiens, France

Bachelor's Degree in Computer Science & Networks *(Honors)* **09/2020 – 09/2023**

- Specialization: IoT Devices and Industrial Systems.
- Core Modules: Computer Science Fundamentals, Algorithms, Databases, AI & Computer Arithmetic.

Experience

Aumovio/Continental Automotive, Apprenticeship, France & Hungary

Embedded Systems Engineer " Sensor Fusion " – Budapest **(Upcoming) 04/2026 – 09/2026**

- Selected to join a mixed Agile team of engineers and mathematicians.
- Technical focus: C++ and FPGA development.
- Objective: Implementation of multi-sensor data fusion algorithms for autonomous driving systems.

Product Owner/Validation Engineer – Toulouse **01/2025 – 04/2026**

- Co-managed a multi-site project involving 20-person team for **Aurora Innovation** (client).
- Designed an automated verification system (Hardware/Software test benches) for end-of-line production.
- Balanced role (50/50) between technical leadership and project organization.

Low-Level Software Engineer (HPC) – Toulouse **10/2023 – 01/2025**

- Integrated into a specialized 12-person Agile team focusing on Embedded Linux.
- Worked on Software Defined Vehicle (SDV) architectures using High-Performance Computing (HPC).

NXO (ex-Nextira One), Apprenticeship, Amiens, France

Expert On Site **10/2022 – 09/2023**

- Deployed and maintained network systems and datacenters for major industrial clients (Goodyear, Lactalis).

University of Technology of Compiègne (UTC), Internship, Compiègne, France

Software Developer Intern **03/2022 – 06/2022**

- Developed internal software tools in collaboration with the university IT department.

Skills

- Hardware & Design: FPGA (Xilinx), Synthesis, VHDL, CPU Architecture (RISC/CISC), Microcontrollers (STM32, ARM Cortex, Automotive-specific: Stellar, S32GX), Microelectronics.
- Programming: C/C++ (Embedded & High Level), Python (Scripting & Data), Rust, C#.
- Tools & Environment: Vivado, Git, Linux (Shell/Bash), CI/CD, Agile Methodologies (Scrum).
- Languages: English (C1 – Professional Proficiency, TOEIC: 970/990), French (Native).

Interests

- *Memberships*: Member of the SIA (French Society of Automotive Engineers).
- LaTeX Package Creator: Created and currently maintaining the callouts-box package on CTAN (Comprehensive TeX Archive Network) - https://ctan.org/pkg/callouts-box
- Sports: Motorcycle and Enduro riding